



Amr Aboughazala

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Objective

I'm an algorithm and data processing engineer with experience in signal processing and software development. My passion lies in transforming mathematical concepts into practical code and real-time solutions, from optimization methods to intuitive tools that make complex systems more accessible.

Industry Experience



Sep. 21
Present

Senior Algorithm and Data Processing Developer, Scantinel Photonics GmbH, Ulm Germany

- Developed a **novel algorithm** addressing Doppler ambiguity in FMCW LiDAR, improving velocity estimation for fast-moving nearby targets. Starting with research on recent publications into practical implementations with CARLA-based simulation and deployment on real-time embedded systems.
- Implemented an internal **perception pipeline** with point cloud **novel filtering** algorithms, Ransac segmentation, kNN, KD-tree for object detection and Kalman tracking. Implemented PointNet and PV-RCNN in PyTorch, building a Docker-based development environment.
- Developed **pixel-level GUI statistical analysis tools** to evaluate different filtering and detection algorithms on raw time-base data (Hann, Chebyshev, CFAR, RANSAC, M-estimator).
- Applied Kalman and Extended Kalman Filters for object tracking within LiDAR perception pipelines; familiar with Unscented Kalman Filtering and probabilistic state-estimation techniques.
- Developed and optimized real-time LiDAR processing algorithms for ARM-based embedded systems, ensuring reliable performance between Scan Controller, ARM processor, and host PC via UDP/TCP with PTP synchronization under resource and timing constraints.
- Designed and implemented the LiDAR system software architecture using MVC in PyQt, integrating real-time and offline visualization tools.



Jan. 19
Jul. 21

Software Developer, Navimatix GmbH, Jena Germany

- Implemented an image processing pipeline for **bacteria counting** on fluorescence microscopy images (E. coli, P. aeruginosa, M. tuberculosis, S. aureus), integrating ImageJ/FIJI APIs in a JavaFX GUI.
- The workflow applied Median filtering, ISODATA segmentation, Opening filters, and 2D/3D object counting using convex-acute angles to separate overlapping Bacteria.
- Implemented several user interface applications in Java (JavaFX), C# (.Net Framework WPF) and Delphi.



Sep. 16
Feb. 17

Working Student, Siemens, Network R&D, Munich Germany

- Implemented a simulation of the IEEE Time-Sensitive Networking (TSN) Scheduler using Omnet++ with C++ 13.
- Developed a C++ parser to integrate SDN Controller output topology into the omnet++ simulator, enabling real time analysis of chosen topology.



Jul. 11
Aug. 14

Customer Technical Network Specialist, Orange Business Services, Cairo Egypt

- Diagnosed and resolved WAN network faults (proactive and reactive), working extensively with IPv4/IPv6, MPLS, and routing protocols including OSPF, RIP, EIGRP, and BGP, ensuring SLA compliance and service continuity.
- Shift Leader for a group of 5 to 10 daily, managing the workflow through the team.

Education



Sep. 14
Feb. 18

M.Sc. in Communications and Signal Processing, Technische Universität Ilmenau, Germany

- Major Subjects: Mobile Communications Engineering, Adaptive Array and Signal Processing, Digital Signal Processing (audio and Image) and Communication Networks.
- Average grade: 2.2/1.0
- Master Thesis 'Positivity Decomposition Algorithms on EEG/MEG Data': Implemented non-negativity constraints on tensor-based blind source separation algorithm
Sources: [Publication](#), [Thesis Publication \(not completed\)](#)
- Advanced Research Project 'Decomposition of a Low Rank Tensor with Missing Entries': Developed a missing imputation tensor algorithm to make it adaptive as per step size and rank estimation.
[Sources](#)



Sep. 05
Feb. 10

B.Sc. in Electronics and Communication, Arab Academy for Science and Technology, Alexandria

- Major Subjects: Wireless and Mobile Communications, Analog and Digital Signal Processing, Communication Networks, Electromagnetic and Antenna theories.
- Average grade: 1.3/1.0 – Excellent with degree of honor besides graduating top of my class.

Certificates and Online Courses

- 22 May 2019: [OOP in Java — Udemy](#)
- 24 Aug. 2020: [Machine Learning — Coursera](#)
- 14 May 2021: [Structuring ML Projects — Coursera](#)
- 30 Apr. 2021: [Neural Networks and Deep Learning — Coursera](#)
- 04 Jun. 2021: [Convolutional Neural Networks — Coursera](#)
- 07 May 2021: [Improving DNN Hyperparameter Tuning, Regularization and Optimization — Coursera](#)
- 24 Aug. 2021: [Sensor Fusion — Udacity \(Mercedes Benz\)](#)
- 02 Sep. 2025: [Introduction to Deep Learning & Neural Networks with Keras — Coursera \(IBM\)](#)

Technical Skills

Coding

Matlab ●●●●●●
Python ●●●●●●
C++ ●●●●●●
Java ●●●●●●
C# ●●●●●●

DATA: numpy, pandas, scipy
PLOT: matplotlib, plotly, pyqt-graph
ML: pytorch, scikit-learn, spconv
CV: opencv

Libraries

VC: git, gitlab
GUI: PyQt, JavaFX, WPF
OOP: MVC, MVVM
LIDAR: open3D, OpenMMLab, Open3D-ML, pytorch3d

Technical Fields

Signal Processing
[Optimization-Mathematics](#)
Machine Learning
Wireless Communication
Audio & Image Processing
Communication Networks

Personal Skills

Languages

Arabic: Native
English: Fluent
German: B1 TELC
Hören ●●●●●●
Lesen ●●●●●●
Schreiben ●●●●●●
Sprechen ●●●●●●

Interests

Graphics Design
Tennis
Movies and TV Series
Trying Different Restaurants